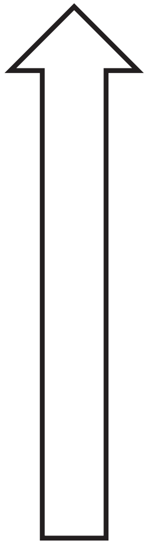
6.4.7 Wrap-Up: Reflection

1. Color in the arrow up to the statement that best describes your current level of confidence with using a number line to add integers.



I'm very confident! I can teach anyone how to add integers.

I can help others add integers.

I can add integers on my own.

I can add integers with assistance.

I'm having trouble adding integers.

I'm not sure what integers are.

Unit 6, Lesson 5: Adding Integers

Benchmark

MA.6.NSO.4.1 Apply and extend previous understandings of operations with whole numbers to add and subtract integers with procedural fluency.

Learning Target

☐ I can add integers with procedural fluency.

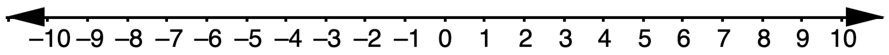
6.5.1 Warm-Up: True Equations

1. Using integers between -3 and 3 one time each, fill in the blanks to make the equation true.

$$\square + \square = \square$$

6.5.2 Exploration: Additive Inverses

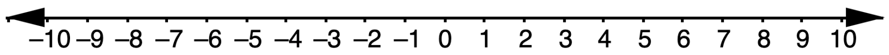
1. Use the number line to show the sum of $-4+4$. Then, complete the equation.



$$-4+4= \_\_\_\_\_\_$$

2. Represent the sum of $-4+4$ using integer chips.

3. Use the number line to show the sum of $7+(-7)$. Then, complete the equation.



$$7+(-7)\ = \_\_\_\_\_\_$$

4. Represent the sum of $7+(-7)$ using integer chips.

Existence of Additive Inverses

For every a there exists $(-a)$ so that $a+(-a)=0$ and $(-a)+a=0$.

Additive inverses are another way to represent the concept of a zero pair.

5. What number is the additive inverse of -15?

6. What number is the additive inverse of 27?

7. Choose any positive integer and write it in the blank: _____.

   Use your number to fill in the blanks in the statement.

   For the value _____, there exists $(-$_____$)$ so that _____ $+(-$_____$)=0$ and $(-$_____$)+$ _____ $=0$.

8. Write your partner's positive integer: ______.

   Use your partner's number to fill in the blanks in the statement.

   For the value _____, there exists $(-$_____$)$ so that _____ $+(-$_____$)=0$ and $(-$_____$)+$ _____ $=0$.

6.5.3 Guided Instruction: Adding Integers

Understanding the existence of additive inverses can be helpful when adding integers with larger absolute values.

1. Complete each statement based on your understanding of adding integers so far.

   a. When adding two positive integers, the sum is
      - ○ positive.
      - ○ negative.

   b. When adding two negative integers, the sum is
      - ○ positive.
      - ○ negative.

When adding a positive and a negative integer, the sum will be either positive or negative, depending on which value has the larger absolute value.

2. Consider the expression $40+(-165)$.

   a. What will be the sign of the sum?
      - ○ positive
      - ○ negative

   b. Explain your reasoning.

When adding these integers, zero pairs are created.

c. Use additive inverses to explain how many zero pairs are created.

d. What remains after combining the additive inverses?

e. Complete the equation.

$$40 + (-165) = \_\_\_\_\_\_$$

3. Complete the statements to describe how to find the sum of $-1{,}029 + (-38)$.

When adding $-1{,}029 + (-38)$, [○ zero pairs / ○ no zero pairs] are created because the signs of the addends are [○ the same. / ○ opposites.] The sum can be found by adding the absolute values of the addends. Since both addends are [○ positive, / ○ negative,] the ray for the second addend of the sum will go farther to the [○ right (→) / ○ left (←)] of zero on a number line.

The sum of $-1{,}029 + (^{-}38)$ is [○ 991. / ○ -991. / ○ 1067. / ○ -1067.]

6.5.4 Collaboration: Reasoning With Integers

1. Work with your partner to fill in the blanks with an integer to make each equation true.

$$\boxed{14} + \boxed{} = \boxed{-35}$$

$$\boxed{} + \boxed{-70} = \boxed{234}$$

$$\boxed{-136} + \boxed{} = \boxed{25}$$

2. Choose one equation and explain your thinking in determining the missing value.

6.5.5 Your Turn!

1. Find each sum.

   a. $10 + (-204)$

   b. $(-99) + (99)$

   c. $904 + (-11)$

   d. $-1 + 76$

   e. $205 + (-643)$

   f. $-151 + (-70)$

   g. $-399 + (-405)$

   h. $-80 + 606$

   i. $703 + (-190)$

   j. $-600 + (-78)$

6.5.6 Wrap-Up: Growth Mindset

1. Today, I am still unsure about ______________________

   ____.

2. To fill in this gap, I intend to ______________________

   ____.